

D691

Elementary Mathematics Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A state park in a fourth-grade textbook recorded 47,382 visitors last summer, and its tallest tree is 250 feet tall. What is the value of the digit 7 in the visitor count?

- A. 7
- B. Thousands
- C. 7,000
- D. 700

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Two third-grade classes at a school with a new gym compare how much of their pizzas they ate at a party. Class A ate $\frac{3}{8}$ of a pizza and Class B ate $\frac{2}{5}$ of a same-size pizza. Class A has 24 students. Which class ate more?

- A. Class A, because 3 and 8 are larger numbers than 2 and 5.
- B. Class B, because $\frac{2}{5} = \frac{16}{40}$ is greater than $\frac{3}{8} = \frac{15}{40}$.
- C. Class A, because it has 24 students.
- D. They ate the same amount.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which fraction is greater than $\frac{3}{4}$?

- A. $\frac{3}{5}$
- B. $\frac{4}{5}$
- C. $\frac{2}{3}$
- D. $\frac{5}{8}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A student who loves puzzles solves 6×13 in her head during a math talk on a Friday afternoon by thinking " $6 \times 10 = 60$ and $6 \times 3 = 18$, so 78." Which property did she use?

- A. The associative property.
- B. The commutative property.
- C. The distributive property.
- D. The identity property.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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5.

A school with an award-winning garden club is fencing a rectangular garden 8 meters long and 5 meters wide. The club has 30 members. How many meters of fence are needed to go all the way around?

- A. 26 m
- B. 13 m
- C. 40 m
- D. 30 m

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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6. [TRANSFER — reversed stem / different variable]

A rectangle has a perimeter of 24 cm and a length of 8 cm. What is its area?

- A. 192 cm²
- B. 64 cm²
- C. 24 cm²
- D. 32 cm²

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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7.

A student at a school with brand-new geometry tiles holds up a square tile. The tile is blue and 5 cm on each side. Which statement about the square is true?

- A. It is also a rectangle, a rhombus, and a parallelogram.
- B. It is a trapezoid but not a parallelogram.
- C. It is a rhombus only because it is blue.
- D. It is a square, so it cannot be a rectangle.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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8.

A fifth-grade class in a school with a new library records how many books five students read over break: 3, 4, 4, 5, and 24. The student who read 24 won a reading award. Which measure best describes a typical student, and what is it?

- A. The mean, 8 books.
- B. 24 books, since the award winner sets the standard.
- C. The range, 21 books.
- D. The median, 4 books.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A bag in a third-grade math center, which sits next to the class fish tank, holds 3 red, 5 blue, and 2 green cubes. A student pulls one cube without looking. What is the probability it is blue?

- A. $\frac{5}{3}$
- B. $\frac{1}{3}$
- C. $\frac{1}{2}$
- D. 1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A spinner has 12 equal sections. The probability of landing on red is $\frac{1}{4}$. How many sections are red?

- A. 9
- B. 3
- C. 4
- D. 48

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A fourth grader's bean plant, grown in a clay pot her grandmother painted, is 3 feet 4 inches tall. The class chart lists heights in inches. How tall is the plant in inches?

- A.** 36 inches
- B.** 34 inches
- C.** 7 inches
- D.** 40 inches

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A ribbon is 450 centimeters long. How long is it in meters?

- A.** 4.5 m
- B.** 45 m
- C.** 45,000 m
- D.** 0.45 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A state park in a fourth-grade textbook recorded 47,382 visitors last summer, ~~and its tallest tree is 250 feet tall~~. What is the value of the digit 7 in the visitor count?

Target word: value of a digit **Decides:** The 7 is in the thousands place.

A	Dropped element	The digit, without its place value.
B	Adjacent category	Names the place, not the value.
C	Correct	$7 \times 1,000 = 7,000$.
D	Confusion set	Hundreds place; the 7 is one place further left.

2. Answer B Concept 2

Two third-grade classes ~~at a school with a new gym~~ compare how much of their pizzas they ate at a party. Class A ate $\frac{3}{8}$ of a pizza and Class B ate $\frac{2}{5}$ of a same-size pizza. ~~Class A has 24 students~~. Which class ate more?

Target word: compare fractions **Decides:** $\frac{3}{8} = \frac{15}{40}$ and $\frac{2}{5} = \frac{16}{40}$.

A	Confusion set	Treats fractions like whole numbers.
B	Correct	Common denominator 40 shows $\frac{2}{5}$ is larger.
C	Camouflage	Class size does not change how much pizza was eaten.
D	Dropped element	Drops the comparison step; $\frac{15}{40} \neq \frac{16}{40}$.

3. Answer B Concept 2 · transfer

Which fraction is greater than $\frac{3}{4}$?

Target word: find a greater fraction **Decides:** Compare to 0.75.

A	Inverted operation	Same numerator with a larger denominator — smaller pieces, so 0.6.
B	Correct	$\frac{4}{5} = 0.8 > 0.75$.
C	Adjacent category	≈ 0.667 , close but less than $\frac{3}{4}$.
D	Confusion set	0.625; a bigger denominator does not mean a bigger fraction.

4. Answer C Concept 3

A student ~~who loves puzzles~~ solves 6×13 in her head during a math talk ~~on a Friday afternoon~~ by thinking " $6 \times 10 = 60$ and $6 \times 3 = 18$, so 78." Which property did she use?

Target word: name the property **Decides:** She split 13 into $10 + 3$ and multiplied each part.

A	Confusion set	Associative regroups factors, like $(2 \times 3) \times 4 = 2 \times (3 \times 4)$.
B	Adjacent category	Would be $13 \times 6 = 6 \times 13$ — changing order.
C	Correct	$6 \times (10 + 3) = 6 \times 10 + 6 \times 3$.
D	Inverted operation	Multiplying by 1 leaves a number unchanged; not used here.

5. Answer A Concept 4

A school with an award-winning garden club is fencing a rectangular garden 8 meters long and 5 meters wide. ~~The club has 30 members.~~ How many meters of fence are needed to go all the way around?

Target word: perimeter **Decides:** Fence goes around: $2 \times (8 + 5)$.

A	Correct	$8 + 5 + 8 + 5 = 26$.
B	Dropped element	Adds only two sides.
C	Confusion set	That is the area, in square meters.
D	Wrong stem number	The number of club members.

6. Answer D Concept 4 • transfer

A rectangle has a perimeter of 24 cm and a length of 8 cm. What is its area?

Target word: solve for width, then area **Decides:** Width = $(24 - 16) \div 2 = 4$.

A	Inverted operation	Multiplies perimeter by length.
B	Adjacent category	Treats it as a square with side 8.
C	Wrong stem number	The perimeter value.
D	Correct	Width 4 cm; $8 \times 4 = 32$.

7. Answer A Concept 5

A student at a school ~~with brand new geometry tiles~~ holds up a square tile. ~~The tile is blue and 5 cm on each side.~~ Which statement about the square is true?

Target word: classify a square **Decides:** Categories nest; a square meets the definitions of rectangle, rhombus, parallelogram.

A	Correct	It has four right angles, four equal sides, and two pairs of parallel sides.
B	Confusion set	Squares have two pairs of parallel sides — a parallelogram.
C	Camouflage	Color has nothing to do with classification.
D	Absolute language	Treats categories as exclusive.

8. Answer D Concept 6

A fifth-grade class ~~in a school with a new library~~ records how many books five students read over break: 3, 4, 4, 5, and 24. ~~The student who read 24 won a reading award.~~ Which measure best describes a typical student, and what is it?

Target word: typical value with an outlier **Decides:** 24 is an outlier; the median resists it.

A	Confusion set	$(3 + 4 + 4 + 5 + 24) \div 5 = 8$, pulled up by the outlier.
B	Camouflage	Keys on the award; one extreme value is not typical.
C	Adjacent category	Range measures spread, not a typical value.
D	Correct	Middle of 3, 4, 4, 5, 24 is 4; the outlier does not move it.

9. Answer C Concept 7

A bag in a third-grade math center, ~~which sits next to the class fish tank~~, holds 3 red, 5 blue, and 2 green cubes. A student pulls one cube without looking. What is the probability it is blue?

Target word: P(blue) **Decides:** 5 blue out of 10 total.

A	Wrong stem number	Divides blue by red count.
B	Adjacent category	Counts colors (1 of 3), not cubes.
C	Correct	$5 \div 10 = 1/2$.
D	Inverted operation	Divides blue by non-blue ($5 \div 5$).

10. Answer B Concept 7 · transfer

A spinner has 12 equal sections. The probability of landing on red is $1/4$. How many sections are red?

Target word: solve for favorable outcomes **Decides:** $12 \times 1/4$.

A	Confusion set	The number of non-red sections.
B	Correct	$12 \times 1/4 = 3$.
C	Wrong stem number	The denominator of the probability.
D	Inverted operation	Multiplies 12 by 4 instead of dividing.

11. Answer D Concept 8

A fourth grader's bean plant, ~~grown in a clay pot her grandmother painted~~, is 3 feet 4 inches tall. The class chart lists heights in inches. How tall is the plant in inches?

Target word: convert feet and inches to inches **Decides:** $3 \times 12 + 4$.

A	Dropped element	Converts the feet but drops the 4 inches.
B	Confusion set	Writes the feet and inches side by side as digits.
C	Inverted operation	Adds $3 + 4$ without converting.
D	Correct	$3 \times 12 = 36$; $36 + 4 = 40$.

12. Answer A Concept 8 · transfer

A ribbon is 450 centimeters long. How long is it in meters?

Target word: convert to a larger unit **Decides:** Smaller to larger: divide by 100.

A	Correct	$450 \div 100 = 4.5$.
B	Dropped element	Divides by 10, dropping a place.
C	Inverted operation	Multiplies instead of dividing.
D	Adjacent category	Divides by 1,000, the millimeter-to-meter factor.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Place value	A digit's value is the digit times its place (ones, tens, hundreds, thousands...); each place is ten times the place to its right.	Giving the digit or the place name instead of its value.	"What is the value of the digit...?"
2	Comparing fractions	Compare fractions by finding a common denominator or converting to decimals; a larger denominator means smaller pieces.	Thinking the fraction with the bigger numbers is bigger.	Two fractions with different denominators.
3	Properties of operations	Commutative changes order, associative changes grouping, distributive splits a factor across a sum: $a \times (b + c) = a \times b + a \times c$.	Mixing up associative and distributive, or multiplying only the first addend.	A student's mental-math steps to be named or checked.
4	Area vs. perimeter	Area is square units covering the inside (length $\times$ width for rectangles); perimeter is the distance around (sum of all sides).	Swapping the formulas or adding only two sides.	"How much fence?" vs. "how much carpet?"
5	Classifying quadrilaterals	Categories nest: every square is a rectangle and a rhombus; every rectangle and rhombus is a parallelogram.	Treating categories as exclusive ("a square is not a rectangle").	"Which name also describes...?"
6	Mean vs. median with outliers	The median is the middle value and resists outliers; the mean is pulled toward extreme values.	Reporting the mean as "typical" when one value is extreme.	A data set with one value far from the rest.
7	Simple probability	Theoretical probability = favorable outcomes $\div$ total equally likely outcomes.	Dividing favorable by unfavorable outcomes.	Counts of colors, sections, or cards.
8	Converting measurement units	Going from a larger unit to a smaller one, multiply; from smaller to larger, divide (1 ft = 12 in; 1 m = 100 cm).	Multiplying when converting to a larger unit.	A measurement to be rewritten in another unit.

Course scope (public description): Elementary Mathematics Curriculum is a detailed and practical course designed for educators who seek to expand their knowledge of the mathematics curriculum in elementary classrooms. This course provides the essential mathematical concepts suitable for young learners, including numbers, operations, and algebraic thinking, spanning through the domains of geometry, measurement, data, statistics, and probability. Emphasis is placed on engaging with a curriculum that not only aligns with educational standards but also nurtures a love for mathematics. By the end of this course, candidates are provided with the tools and confidence to foster a strong mathematical foundation in their elementary students.